```
def fractional_knapsack(weights, profits, capacity):
    ratio = [(profits[i] / weights[i], weights[i], profits[i])
             for i in range(len(weights))]

    ratio.sort(reverse=True, key=lambda x: x[0])

    total_profit = 0
    selected_items = []

    for r, w, p in ratio:
        if capacity <= 0:
            break

        if w <= capacity:
            capacity -= w
            total_profit += p
            selected_items.append((w, p, 1.0))
        else:
            fraction = capacity / w
            total_profit += p * fraction
            selected_items.append((w, p, fraction))
            capacity = 0

    return total_profit, selected_items



def input_data():
    n = int(input("Enter number of parcels: "))
```

```python
    weights = []
    profits = []

    for i in range(n):
        w = float(input(f"Enter weight of parcel {i+1}: "))
        p = float(input(f"Enter profit of parcel {i+1}: "))

        weights.append(w)
        profits.append(p)

    capacity = float(input("Enter truck capacity: "))

    return weights, profits, capacity


def display_result(selected, max_profit):
    print("\nSelected parcels:")

    for w, p, frac in selected:
        print(f"Weight: {w}, Profit: {p}, Taken: {frac*100:.2f}%")

    print(f"\nMaximum Profit: {max_profit:.2f}\n")


# Main menu loop
while True:
    print("\n=== Fractional Knapsack Menu ===")
    print("1. Enter Data and Calculate")
    print("2. Exit")
```

```python
    choice = input("Enter choice: ")

    if choice == "1":
        weights, profits, capacity = input_data()

        max_profit, selected = fractional_knapsack(
            weights, profits, capacity
        )

        display_result(selected, max_profit)

    elif choice == "2":
        print("Exiting... Goodbye!")
        break

    else:
        print("Invalid choice. Try again.")
```

FIE NAME

fractional_knapsack.py

COMMAND

python fractional_knapsack.py